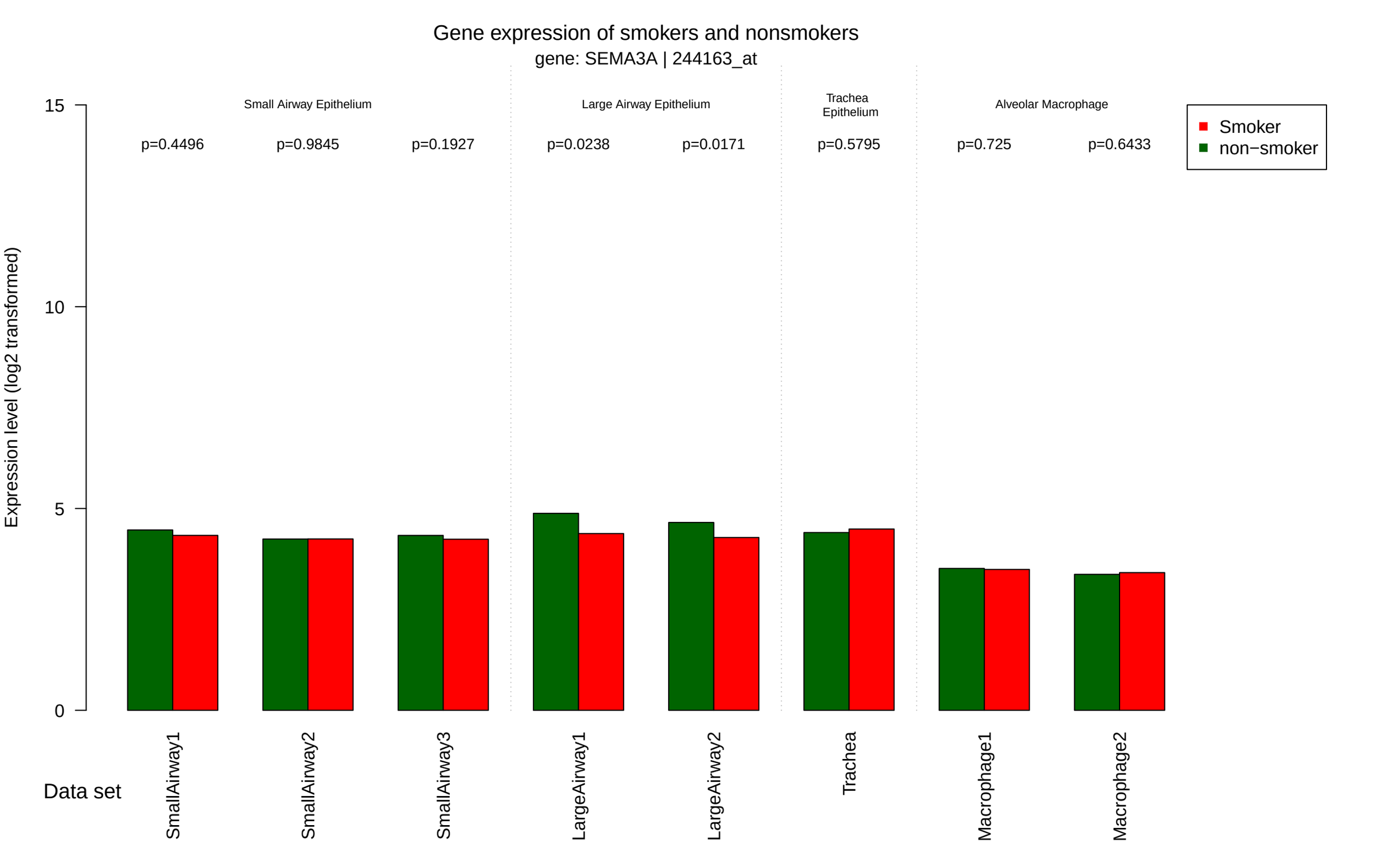
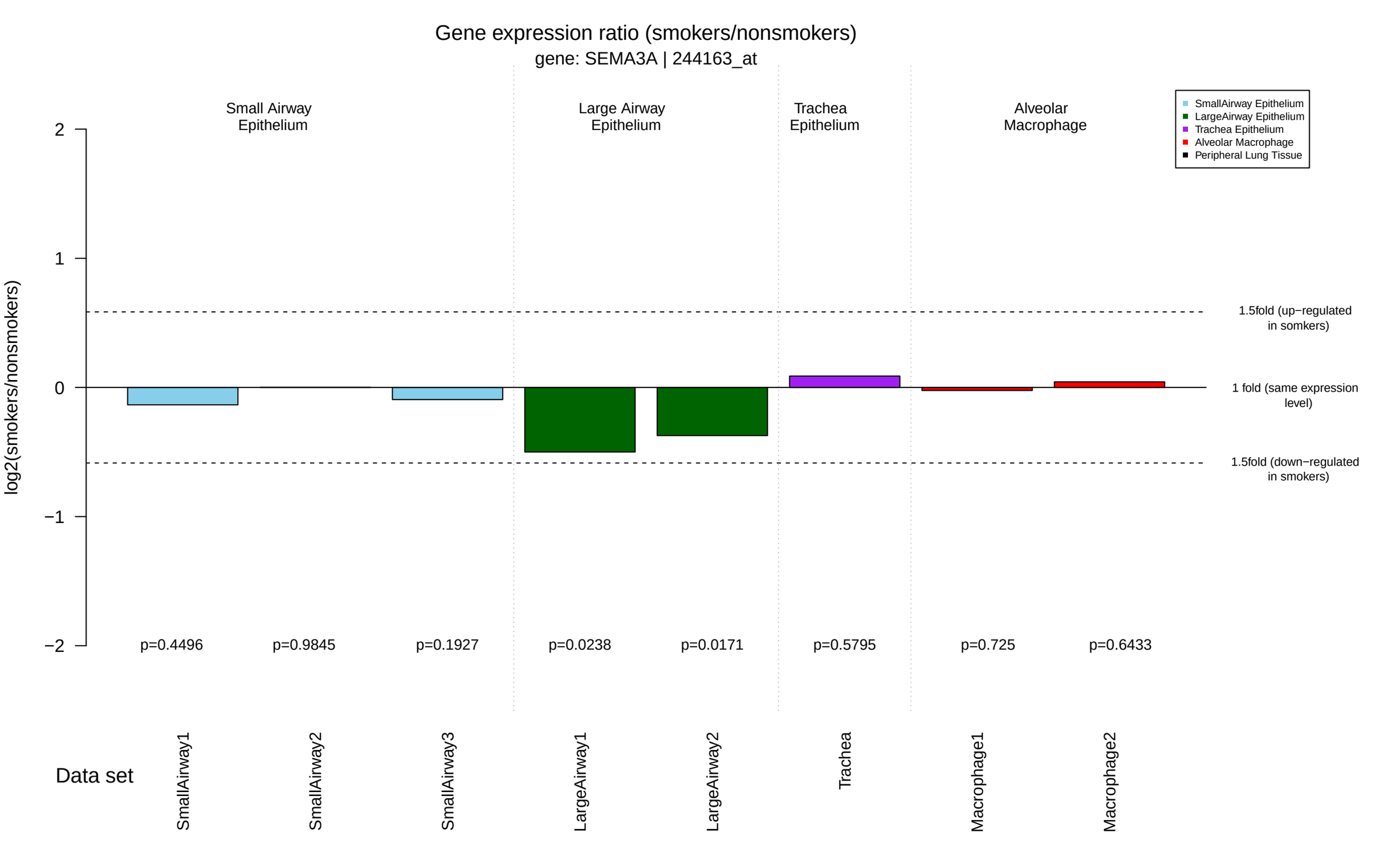
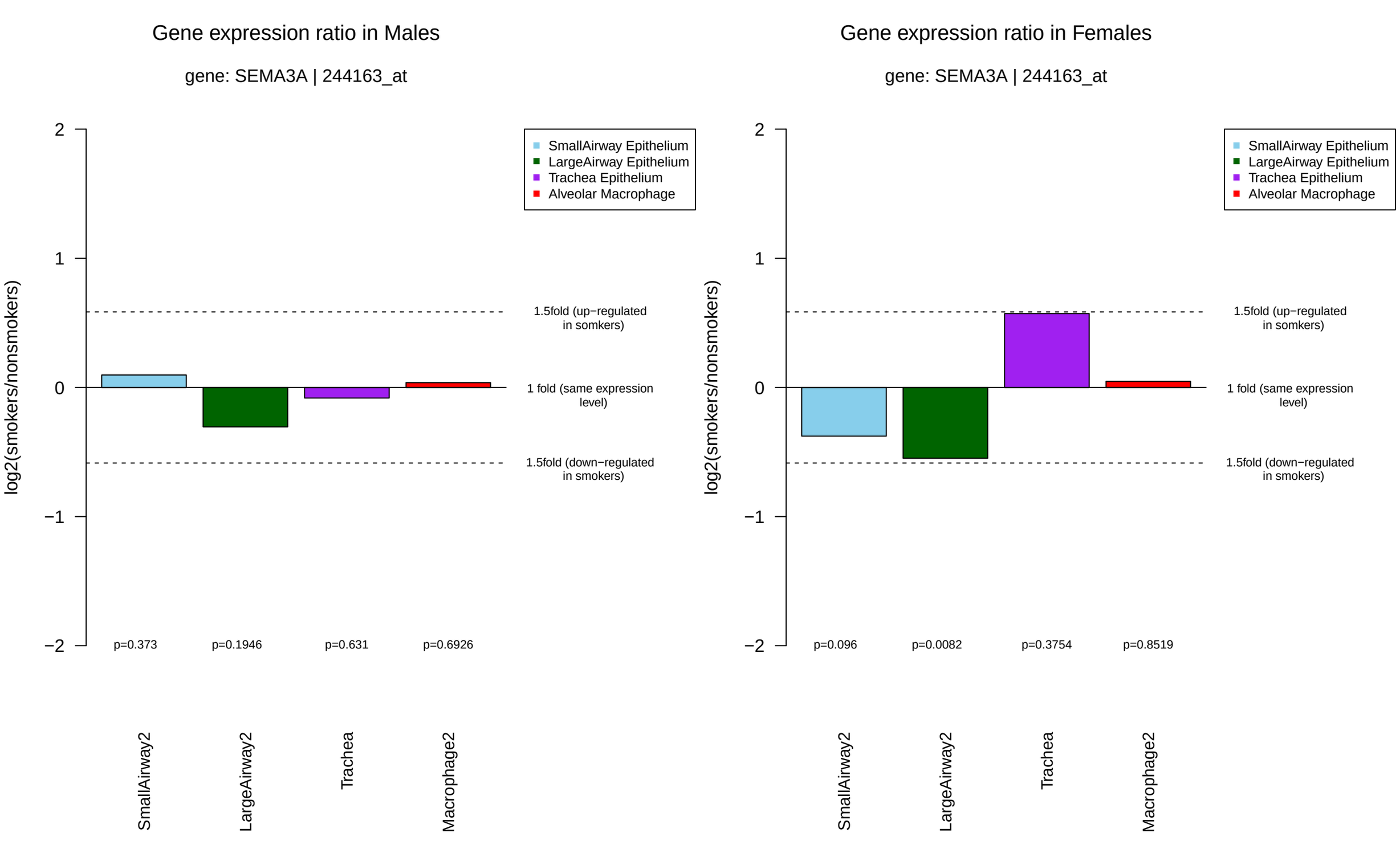
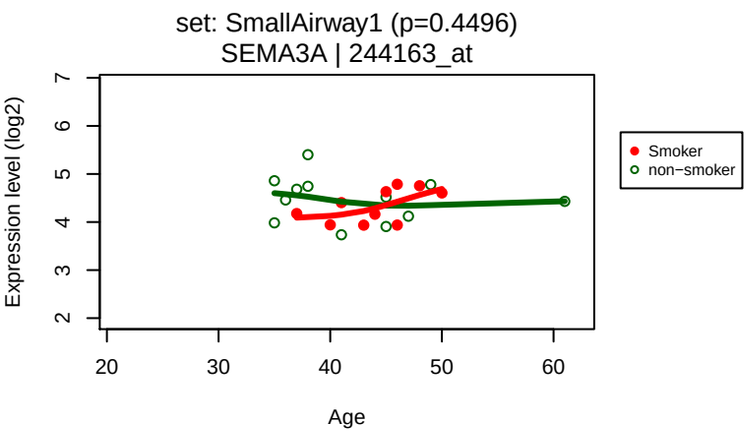
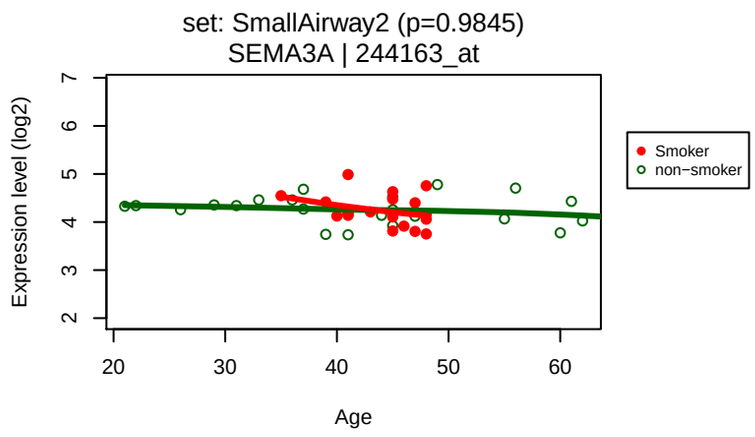
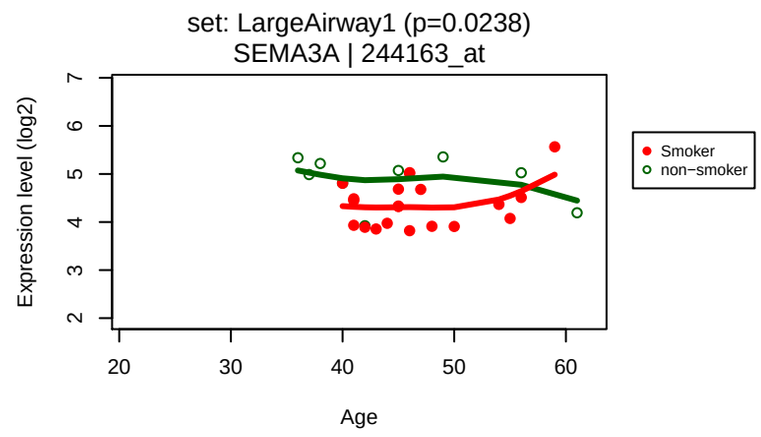

Gene expression of smokers and nonsmokers
gene: SEMA3A | 244163_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4496
p=0.9845
p=0.1927
p=0.0238
p=0.0171
p=0.5795
p=0.725
p=0.6433
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

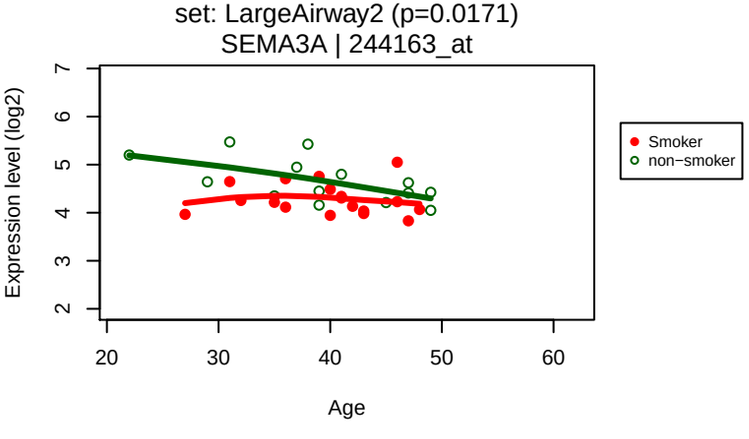

Gene expression ratio (smokers/nonsmokers)
gene: SEMA3A | 244163_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4496
p=0.9845
p=0.1927
p=0.0238
p=0.0171
p=0.5795
p=0.725
p=0.6433
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

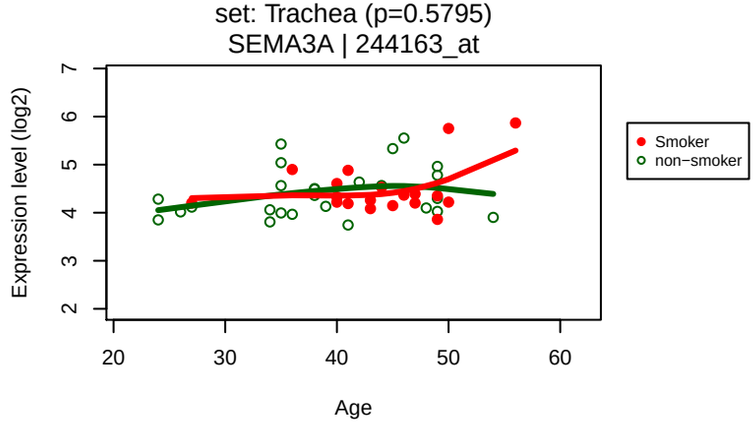

Gene expression ratio in Males
gene: SEMA3A | 244163_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.373
p=0.1946
p=0.631
p=0.6926
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SEMA3A | 244163_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.096
p=0.0082
p=0.3754
p=0.8519
SmallAirway2
LargeAirway2
Trachea
Macrophage2

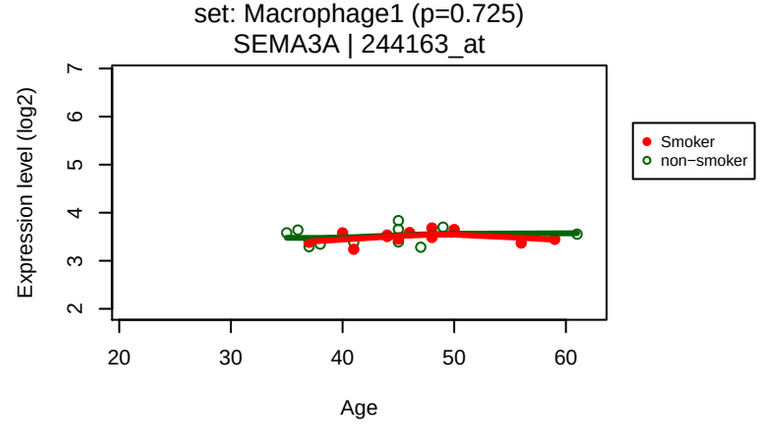

set: SmallAirway1 (p=0.4496)
SEMA3A | 244163_at
Expression level (log2)
Age
Smoker
non−smoker

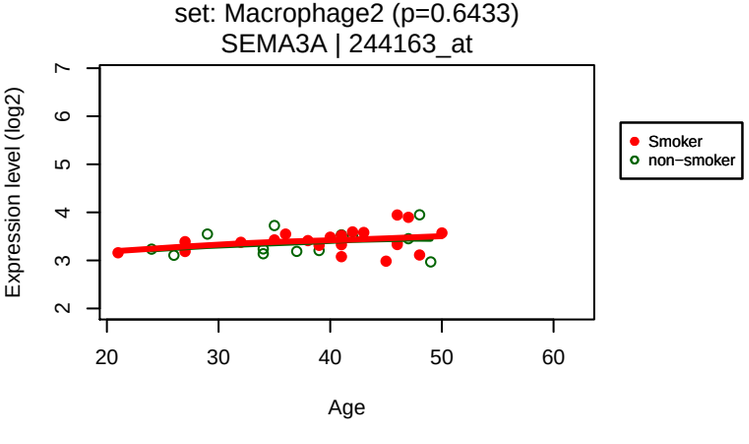

set: SmallAirway2 (p=0.9845)
SEMA3A | 244163_at
Expression level (log2)
Age
Smoker
non−smoker

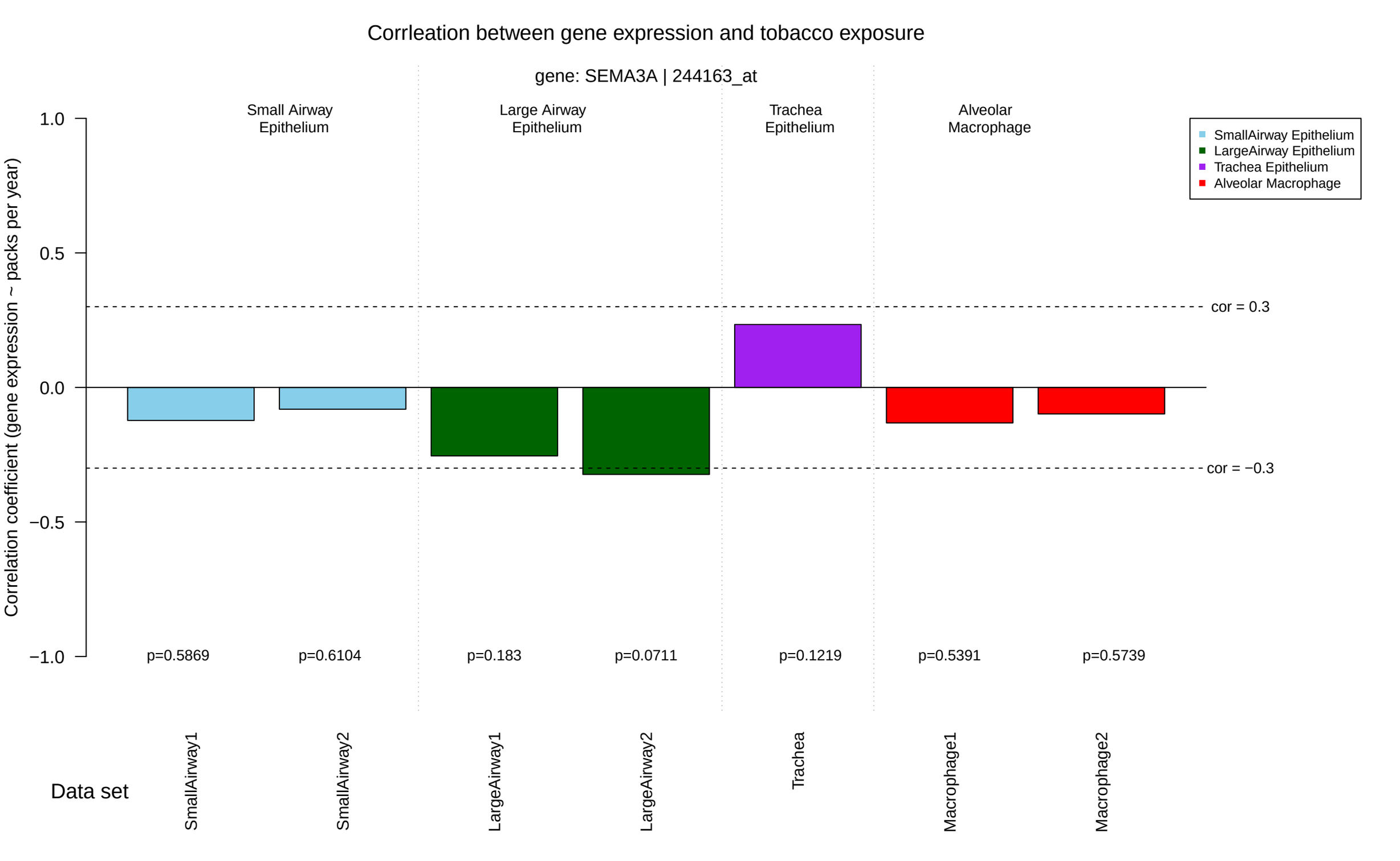

set: LargeAirway1 (p=0.0238)
SEMA3A | 244163_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0171)
SEMA3A | 244163_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5795)
SEMA3A | 244163_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.725)
SEMA3A | 244163_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6433)
SEMA3A | 244163_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SEMA3A | 244163_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5869
p=0.6104
p=0.183
p=0.0711
p=0.1219
p=0.5391
p=0.5739
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2